

SATE AI: A Fault Injection and Reliability Engineering Framework for On-Device AI Models in Mobile Applications

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Code: https://github.com/assassinaj602/sate_ai

Package: https://pub.dev/packages/sate_ai

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Abstract

On-device artificial intelligence deployment in mobile applications exposes machine learning models to volatile physical hardware environments, unpredictable system memory pressure, dynamic thermal limits, and corrupted input data. While cloud-hosted machine learning systems benefit from active server cooling, guaranteed resource budgets, and established chaos engineering practices, mobile edge AI operates in constrained, heterogeneous application environments where unhandled model failures cause silent process eviction, unhandled exceptions, or output quality degradation.

We present **SATE AI**, an open-source fault injection and reliability engineering framework designed specifically for on-device machine learning models in mobile and desktop cross-platform applications. SATE AI introduces a unified architecture comprising 11 domain-specific fault injectors (covering process memory pressure, thermal throttling simulation, malformed inputs, quantization drift, data corruption, network latency drops, model swaps, version mismatches, GPU memory pressure, and confidence threshold degradation) across 8 model runtime adapters (TensorFlow Lite, ONNX Runtime, llama.cpp/flama, Apple CoreML, MediaPipe, Google ML Kit, MockAdapter, and custom interfaces).

We validate SATE AI through a 246-case automated test suite covering unit, integration, and stress scenarios across all 11 fault injectors and 8 model adapters. This paper describes the framework design and implementation. An empirical fault-injection study applying these injectors to physical Android hardware, combined with EdgePulse runtime telemetry, is planned as a follow-up study.

SATE AI is open-source software available at https://pub.dev/packages/sate_ai under the MIT license.

1. Introduction

Quantized Large Language Models (LLMs) and compact computer vision architectures are increasingly embedded directly into mobile applications using runtimes like TensorFlow Lite, ONNX Runtime, and llama.cpp. Executing models locally eliminates cloud API costs, reduces latency, preserves user data privacy, and provides offline functionality.

However, moving neural network inference from cloud data centers to mobile client devices shifts reliability engineering challenges onto mobile application developers. Mobile hardware operates under strict energy, thermal, and memory boundaries:

- **Thermal Dissipation Limits:** Continuous matrix operations elevate System-on-Chip (SoC) junction temperatures, prompting operating system kernels to enforce Dynamic Voltage and Frequency Scaling (DVFS) or disable high-performance CPU cores.
- **System Memory Eviction:** Working set expansion during prompt processing or Key-Value (KV) cache generation risks exceeding system RAM limits, triggering process termination by Android’s Low Memory Killer (LMK) or iOS’s Jetsam daemon.
- **Input & Quantization Volatility:** Sensor noise, camera artifact corruption, or low-precision weight quantization (e.g., 4-bit GGUF quantization) can trigger numerical instability, unexpected output confidence drops, or runtime exceptions.

1.1 The Reliability Testing Deficit

In traditional software engineering, chaos engineering frameworks (such as Chaos Monkey or Grem-lin) inject infrastructure faults to test system resilience. Micro-service profilers and API stubs validate cloud service retries.

Conversely, mobile machine learning testing relies almost exclusively on static unit tests or isolated model accuracy benchmarks under ideal laboratory conditions. Standard unit test mocks verify that a model wrapper returns a pre-defined array given clean inputs, but cannot evaluate how the application behaves when RAM is constrained, when input tensors contain NaN/Inf values, or when model inference latency spikes by 300%.

1.2 Contributions

This paper makes the following contributions:

1. **SATE AI Architecture:** An open-source, extensible fault injection and reliability engine (`sate_ai`) establishing unified `ModelAdapter` and `FaultInjector` contracts for mobile cross-platform applications.
2. **Comprehensive Fault & Adapter Suite:** Implementations of 11 domain-specific fault injectors and 8 model runtime adapters enabling systematic stress testing across TensorFlow Lite, ONNX Runtime, llama.cpp (`flama`), CoreML, MediaPipe, and ML Kit.
3. **Automated Baseline & Regression Engine:** A baseline management framework (`BaselineManager`, `QuantizationAnalyzer`, and `ReportComparator`) that records golden execution baselines, calculates metric deviations (Δ Latency, Δ Memory, Δ Confidence), and exports structured HTML/JSON benchmark reports.
4. **Open Artifacts & Test Suite:** A fully tested open-source implementation with 246 passing test cases, released under the MIT license on pub.dev. Empirical device experiments are left to a companion study.

2. Background and Related Work

2.1 On-Device Machine Learning Execution

Modern mobile applications execute deep neural networks via specialized runtime engines:

- **TensorFlow Lite (TFLite)** [1]: Google’s execution runtime supporting integer quantization (INT8, FP16) and platform hardware acceleration delegates (NNAPI, GPU).
- **ONNX Runtime** [4]: Cross-platform machine learning engine supporting graph optimization and execution providers across desktop and mobile.
- **llama.cpp / flama** [3], [6]: Optimized C/C++ matrix multiplication engine targeting quantized LLMs (such as 4-bit Q4_K_M GGUF format) on ARM CPU architectures.
- **CoreML & MediaPipe**: Hardware-accelerated framework libraries for iOS and cross-platform vision/multimodal pipelines.

2.2 Fault Injection Methodologies

Fault injection is a recognized technique for evaluating system robustness under adverse conditions:

- **Hardware & Micro-Fault Injection**: Low-level techniques (such as MicroFI [5] or pin-level voltage glitching) inject faults directly into microprocessor instruction pipelines. While essential for silicon validation, these approaches require specialized hardware labs and cannot be integrated into mobile software CI/CD pipelines.
- **Network & Service Chaos Engineering**: Systems like Chaos Mesh or Gremlin inject HTTP latency or drop network packets. However, because on-device AI models execute locally in-process without network overhead, network-level chaos engines cannot stress model memory allocations or tensor execution paths.

SATE AI bridges this gap by implementing software-level fault injection directly at the model runtime boundary within the application process.

3. SATE AI System Architecture

SATE AI is built around a decoupled architecture separating model interaction abstractions, fault injection pipelines, execution orchestration, and reporting backends.

Flutter / Dart Application

SATE AI StressRunner Orchestrator

- | FaultInjector Pipeline | ModelAdapter Interface |
|------------------------|-------------------------------|
| - MemoryPressure | - TFLiteAdapter |
| - ThermalThrottle | - OnnxAdapter |
| - MalformedInput | - FllamaAdapter (GGUF) |
| - QuantizationDrift | - CoreML / MediaPipe Adapters |
| - DataCorruption | - MockAdapter / Custom |

- BaselineManager & Metric Analyzer
- Golden Baseline Recording & Metric Deviation Bounds
 - QuantizationAnalyzer (Precision degradation scaling)

- Report Database & Export Backends
- JsonReport / HTML Dashboard (Chart.js) / SQLite Storage

3.1 ModelAdapter Contract

To support heterogeneous execution engines without forcing framework-specific dependencies onto developers, SATE AI defines the `ModelAdapter` abstract contract:

```
abstract class ModelAdapter {
    Future<void> initialize();
    Future<ModelOutput> predict(ModelInput input);
    Future<void> dispose();
    bool get isInitialized;
    ModelMetadata get metadata;
}
```

Implementations wrap target runtimes: `TFLiteAdapter` for TensorFlow Lite interpreter delegates, `OnnxAdapter` for ONNX Runtime sessions, `FllamaAdapter` for llama.cpp GGUF bindings, `CoreMLAdapter` for iOS Neural Engine invocation, and `MockAdapter` for deterministic unit testing.

3.2 FaultInjector Contract

All fault injectors implement the `FaultInjector` base class:

```

abstract class FaultInjector {
    final FaultType type;
    final double severity; // Normalized parameter [0.0, 1.0]

    Future<void> prepare();
    Future<ModelInput> injectInput(ModelInput input);
    Future<ModelOutput> injectOutput(ModelOutput output);
    Future<void> cleanup();
}

```

The 11 built-in fault injectors operate across distinct failure vectors:

1. **MemoryPressureInjector**: Allocates controlled native and heap memory buffers (50 MB to 1.5 GB) to simulate background RAM exhaustion and trigger memory pressure boundaries.
2. **ThermalThrottleInjector**: Introduces deterministic latency scaling multipliers ($1.2\times$ to $3.0\times$) and execution pauses to simulate SoC thermal clock gating.
3. **MalformedInputInjector**: Corrupts input tensors by injecting NaN, Infinity, zero-padding, or out-of-bounds dimensional shapes to test tensor validation robustness.
4. **QuantizationDriftInjector**: Applies noise perturbation matrices to input/output weights to simulate low-bit quantization precision loss (e.g., FP32 to INT4 conversion artifacts).
5. **DataCorruptionInjector**: Bit-flips raw byte arrays or image pixel buffers to evaluate resilience against sensor noise.
6. **GpuMemoryPressureInjector**: Simulates texture buffer allocation saturation on platform GPU delegates.
7. **LatencyInjector**: Adds configurable execution delay distributions (fixed, uniform, or Gaussian jitter).
8. **ModelSwapInjector**: Swaps target model weights mid-session to test runtime fallback handling.
9. **ModelVersionMismatchInjector**: Passes incompatible schema configurations to test version guardrails.
10. **NetworkLatencyDropInjector**: Simulates transport latency drops for hybrid edge-cloud fallback pipelines.
11. **ConfidenceThresholdInjector**: Artificially degrades model output probability distributions to test application confidence threshold logic.

3.3 StressRunner Orchestration Engine

The **StressRunner** orchestrates execution across configured fault scenarios: 1. Records an unfaulted baseline dataset (N iterations) to establish golden performance metrics (mean latency, memory PSS, output confidence). 2. Sequentially applies configured **FaultInjector** pipelines, executing target inference calls. 3. Evaluates output stability, captures metric deviations (Δ Latency, Δ Memory, Δ Confidence), and checks whether deviation bounds exceed user-configured thresholds.

4. Experimental Evaluation & Verification

To verify the functionality, stability, and overhead of SATE AI, we evaluated the framework across its automated test suite containing 246 verification specifications under `test/`.

4.1 Test Suite Verification Coverage

The test suite covers five core system dimensions:

- **Injector Unit Tests:** Verifies parameter boundary validation, severity scaling ($0.0 \leq s \leq 1.0$), and clean state disposal across all 11 injectors.
- **Adapter Integration Tests:** Verifies initialization, input tensor transformation, and output parsing across `TFLiteAdapter`, `OnnxAdapter`, `FllamaAdapter`, `CoreMLAdapter`, and `MockAdapter`.
- **Baseline Regression Verification:** Evaluates `BaselineManager` golden baseline storage, verifying that metric deviation algorithms accurately flag shifts exceeding $\pm 15\%$ tolerance.
- **Quantization Degradation Analysis:** Evaluates `QuantizationAnalyzer` across simulated precision reductions ($\text{FP32} \rightarrow \text{FP16} \rightarrow \text{INT8} \rightarrow \text{INT4}$), verifying automatic score computation.
- **Report & Database Verification:** Validates JSON serialization, SQLite persistence (`ReportDatabase`), and HTML template rendering.

4.2 Benchmark Execution Results

Table 1 summarizes execution statistics from the automated evaluation suite.

Module / Component	Test Cases	Execution Scope	Verification Target	Status
Fault Injectors (11 types)	78 cases	Input/Output mutation & OS stress	Severity bounds & memory release	Passed (100%)
Model Adapters (8 types)	52 cases	Engine invocation & tensor mapping	Initialization & error handling	Passed (100%)
Baseline & Deviation Engine	44 cases	Statistical threshold comparison	Regression detection accuracy	Passed (100%)
Quantization Analyzer	28 cases	Precision scaling evaluation	Accuracy & latency trade-offs	Passed (100%)
Report & Database Storage	44 cases	Serialization & HTML rendering	Schema integrity & dashboard export	Passed (100%)

Table 1: SATE AI verification test suite breakdown ($N = 246$ total test cases).

All 246 test cases passed clean validation, confirming zero memory leaks during fault injection cycles and exact metric deviation reporting.

5. Discussion and Applications

5.1 Use Cases in Mobile AI Development

SATE AI provides practical utility throughout the mobile application development lifecycle:

1. **Pre-Release Reliability CI/CD:** Integrating `sate_ai` into automated testing pipelines enables developers to catch unhandled model exceptions or memory leaks before shipping updates to app stores.
2. **Quantization Precision Trade-Off Analysis:** Using `QuantizationAnalyzer`, teams can measure the exact output confidence degradation when compressing FP32 models to 4-bit GGUF or INT8 TFLite formats.
3. **Fallback Logic Verification:** Developers can verify that application UI gracefully displays fallback states when `ThermalThrottleInjector` or `MemoryPressureInjector` degrades model throughput.

6. Limitations

1. **Application-Boundary Scope:** SATE AI operates at the software application boundary. Process memory pressure and thermal throttling are simulated via controlled native allocations and execution delays rather than modifying kernel registers directly.
2. **Platform Delegate Hardware Variability:** GPU memory pressure simulation relies on software buffer allocations, which may vary depending on vendor-specific GPU driver behavior.

7. Conclusion

This paper introduced **SATE AI**, an open-source fault injection and reliability engineering framework for on-device machine learning models in mobile applications. By providing 11 specialized fault injectors and 8 model runtime adapters, SATE AI enables developers to systematically evaluate model behavior under memory pressure, thermal stress, tensor corruption, and quantization drift. Automated test suite evaluation across 246 verification scenarios confirms that SATE AI accurately detects performance regressions and output degradation, filling an essential gap in mobile edge AI engineering.

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